
Integrable Systems — Exercise Sheet 19

Why integrable systems do not thermalise

Variant: Questions only

Conventions, as in Lecture 19: a ring of N spins $\frac{1}{2}$ and the translation τ ; a local charge is $Q = \sum_n \tau^{n-1}(q)$ for a density q of fixed range r . The quench: Lecture 18's transverse-field Ising chain at coupling J , from its ground state at field h_0 to the field h at $t = 0$, with $a_k = 2(h - J \cos k)$, $b_k = 2J \sin k$, $\varepsilon_k = \sqrt{a_k^2 + b_k^2}$, the angle θ_k with $\cos 2\theta_k = a_k/\varepsilon_k$, $\sin 2\theta_k = b_k/\varepsilon_k$, the same with a superscript 0 at h_0 , and $\chi_k = \theta_k - \theta_k^0$. Lecture 19 proves that the post-quench occupations are $n_k = \sin^2 \chi_k$, and that for $J, h, h_0 > 0$, $h, h_0 \neq J$, $m^z(t) = \int \frac{dk}{2\pi} [\cos 2\theta_k \cos 2\chi_k + \sin 2\theta_k \sin 2\chi_k \cos(2\varepsilon_k t)]$ in the thermodynamic limit, which tends to $m^z(\infty) = \int \frac{dk}{2\pi} \cos 2\theta_k (1 - 2n_k)$. Integrals over k run over $(-\pi, \pi)$.

Exercise 1 (local charges). (i) Show that $\|Q\| \leq N \|q\|$, and that $\langle Q \rangle = N \langle q \rangle$ in a translation-invariant state. (ii) Show that the local charges of range at most 1 form a space of dimension 4, and those of range at most 2 a space of dimension 13 when $N \geq 3$, below Lecture 19's bound $4^2 = 16$. What is it when $N = 2$?

Exercise 2 (dephasing, and a spectator). (i) For $\omega \neq 0$ show that $|\frac{1}{T} \int_0^T e^{-i\omega t} dt| \leq \frac{2}{|\omega|T}$. (ii) Let $H = H_c \otimes \mathbb{1}$, a chain with Hamiltonian H_c and eigenbasis $|n\rangle$, beside a spectator spin that feels nothing, and start from $|\varphi\rangle \otimes |+\rangle$. Show that $\langle \sigma_s^x \rangle(t) = 1$ for all t ; that $\sum |c|^2 \langle \sigma_s^x \rangle$ over the product eigenbasis $|n\rangle \otimes |\uparrow\rangle$, $|n\rangle \otimes |\downarrow\rangle$ is 0; and that Lecture 19's dephased state gives 1. What went wrong with the second?

Exercise 3 (the occupations). (i) Show that $\cos 2\chi_k = \frac{a_k a_k^0 + b_k^2}{\varepsilon_k \varepsilon_k^0}$ and $\sin 2\chi_k = \frac{4J(h_0 - h) \sin k}{\varepsilon_k \varepsilon_k^0}$. (ii) For $J = 1$, $h_0 = 0.4 \rightarrow h = 1.9$, compute $n_{\pi/2}$, and show that n_k is even in k . (iii) Lecture 19 warns that the sum angle $\theta_k + \theta_k^0$ gives a plausible and wrong n_k . For this quench, how wrong, at worst over k ?

Exercise 4 (the generalised Gibbs value, and the Gibbs one). (*Computational.*) Take the quench of Exercise 3. (i) Compute $m^z(\infty)$. (ii) Compute the energy per site $e = \int \frac{dk}{2\pi} \varepsilon_k (n_k - \frac{1}{2})$ after the quench, the temperature T at which the Gibbs state of $H(h)$ has this energy (thermal occupations $1/(e^{\varepsilon_k/T} + 1)$), and the Gibbs value of m^z . (iii) At $h = J$, show that $\cos 2\theta_k = \varepsilon_k/(4J)$, and deduce that $m^z = -e/(2J)$ in every state diagonal in the occupations: there the observable cannot tell the two ensembles apart.

Exercise 5 (the amplitude). (i) Show that $\sin 2\theta_k \sin 2\chi_k = \frac{8J^2(h_0 - h) \sin^2 k}{\varepsilon_k^2 \varepsilon_k^0}$. (ii) For $h, h_0 \neq J$ and $h_0 \neq h$, show that it has a double zero at the stationary points $k = 0, \pi$ of ε_k ; at $h_0 = J$ (with $h \neq J$), only a simple zero in $|k|$ at $k = 0$. What if $h_0 = h$? (Lecture 19's stationary-phase step turns these into decays like $t^{-3/2}$ and t^{-1} .) (iii) What does $m^z(t)$ do at $h = 0$?

Exercise 6 (the ground state is even). Lecture 19's proof of the relaxation theorem needs the pre-quench ground state in the even sector. Take $J, h_0 > 0$. (i) In the basis of joint eigenvectors of every σ_n^x , show that $H(h_0)$ has off-diagonal entries $-h_0$ between configurations differing in one spin, and 0 otherwise. (ii) Perron–Frobenius (an outside fact): a real symmetric matrix whose off-diagonal entries are ≤ 0 , and which connects every basis vector

to every other through non-zero entries, has a non-degenerate lowest eigenvalue with a strictly positive eigenvector. Deduce that the ground state of $H(h_0)$ is unique and, in this basis, positive. (iii) Deduce that it is even.

Exercise 7 ★. *Which charges are local.* For an even f let $Q[f] = \sum_k f(k)(n_k - \frac{1}{2})$, whose real-space hopping and pairing coefficients, in each sector, are the Fourier coefficients of $A = fa_k/\varepsilon_k$ and $B = fb_k/\varepsilon_k$ (Lecture 19's sketch). Take $J, h > 0$ with $h \neq J$. (i) Show that $(h + J \cos k)a_k - J \sin k b_k = 2(h^2 - J^2)$. (ii) Deduce that $Q[f]$ has finite range if and only if $f = \varepsilon \cdot P$ with P a trigonometric polynomial. (iii) (*Computational.*) For $f = 1$ and $h = 0.5J$, compute the ratios $|\hat{A}_{m+1}/\hat{A}_m|$ of consecutive Fourier coefficients of a_k/ε_k at $m = 8, 16, 24, 32$. What do they say about $Q[1]$?